Student Ride-Hailing System

Product requirements for the development team
Initial requirements • Central University, Miotso campus, Prampram, Ghana

1. Product overview

Build a cashless ride-hailing system initially for Central University students at the Miotso campus. Verified students request rides, pay through an in-app wallet, and track their assigned driver. The system comprises a student app and web app, a driver app, and an admin dashboard.

The student experience should feel modern, safe, affordable, and easy to use. The driver app should prioritize clarity and ease of use.

2. Confirmed business decisions

Area	Requirement
Initial users and campus	Central University students at Miotso campus, Prampram.
Verification	Ghana Card and Central University student ID.
Service area	From Central University Miotso toward Ashaiman China Mall, the current cutoff point. Exact boundary remains to be mapped.
Trip limit	Maximum 18 km by road; longer trips cannot be booked.
Payments	Cashless, from funds in the student wallet.
Drivers	Salaried; no commission or percentage of fares.
Assignment	Automatic assignment to an available, on-duty driver.
Cancellations	Free for students; no no-show fee.
Operations	Salary and attendance managed in the admin dashboard.

3. Service area and trip eligibility

The service extends from Central University Miotso toward Ashaiman China Mall, the cutoff point in that direction. Administrators need a configurable map boundary and pickup locations. The exact boundary has not yet been drawn.

- Block bookings when the pickup or destination is outside the configured area or the road route is longer than 18 km. Explain the reason and allow the student to change locations.
- Before launch, check real road distances between the campus and mall against the 18 km limit and approve the exact boundary.

4. Student registration and verification

Students create an account and submit their Ghana Card and student ID. Only verified Central University students may book. Provide pending, approved, needs attention/resubmission, and rejected statuses, with appropriate explanations and notifications.

Administrators need secure review and decision records. Agree the information collected, review process, reviewers, retention periods, and handling of expired or unclear documents. Design access controls, secure storage, and a deletion process before collecting identity documents.

5. Student booking and safety

Students select pickup and destination, see eligibility, road distance, fare and estimated pickup time, then confirm after a wallet balance check. Show the assigned driver, vehicle and live location. Provide a ride PIN for pickup, trip status, a receipt and trip history. Allow free cancellation with a clear confirmation.

Provide trip sharing with a trusted contact, issue and safety reporting, access to support and emergency assistance, and ride ratings and feedback. Agree support and emergency procedures before launch.

6. Fare structure

Use the road-route distance for the individual trip. Show the fare before confirmation. Authorized administrators should be able to configure fare bands.

Road-route distance	Base fare
Up to 3 km	GH₵18
Over 3 km to 6 km	GH₵25
Over 6 km to 12 km	GH₵35
Over 12 km to 18 km	GH₵50
Over 18 km	Booking unavailable

7. Shared rides

- Two students: each pays 50% of the trip fare.
- Three students: each pays 35% of the trip fare plus GH₵5.

Each student pays their own share from their wallet and sees that share before confirming. For a GH₵35 base fare, two students pay GH₵17.50 each; three pay GH₵17.25 each, for a total of GH₵51.75.

Still to agree: Matching and route compatibility; the base fare when riders have different pickup or drop-off points; capacity; what happens when a rider cancels or does not arrive; and how remaining riders approve revised fares. Shared-ride release timing depends on these decisions.

8. Student wallet and payments

Students add funds to a wallet and pay from their balance. Drivers do not collect cash. Show the balance, top-up action, transaction history for top-ups, rides, refunds and adjustments, and ride receipts. Explain insufficient balances before booking.

The intended approach is to check and reserve the required amount at booking, then deduct the final fare at completion. Validate this flow with the selected payment provider.

Still to agree: Top-up methods such as mobile money or bank cards, unused-fund withdrawals, provider fees, failed-payment handling, and optional automatic top-ups. Confirm Ghanaian legal, payment and security requirements with an appropriate provider and adviser.

9. Cancellations, refunds and fare adjustments

All student cancellations are free, including before or after assignment and after driver arrival. There is no no-show fee. Release any reserved amount on cancellation; if a fare was already deducted for a cancelled ride, refund it in full to the wallet. Show the cancellation and refund in the account history. A driver or service cancellation should also incur no student charge.

Proposed rules to confirm: refund incorrect or duplicate charges; let administrators review complaints and safety issues; show and obtain acceptance of a revised fare before a destination change; and block changes outside the area or above 18 km. Decide how to handle partially completed trips and breakdowns.

10. Driver app

Drivers are salaried and receive no percentage of individual fares. They need to sign in; view their profile and vehicle; go on and off duty; receive assigned trips; view and navigate to pickup and destination; mark arrival; enter the student PIN to start; and complete the trip.

Drivers should also be able to report that they cannot take an assignment, see their trips, shifts, attendance and salary payment history, report vehicle or trip issues, contact support, and access emergency assistance. Use large, high-contrast controls and minimize interaction while driving.

11. Automatic driver assignment — key requirement

The system must automatically assign each ride request to an available, on-duty driver. Drivers should not need to compete to accept requests.

- Receive an eligible, confirmed request; find an available, on-duty driver; assign and notify the driver; and show driver and vehicle details to the student.
- Allow the driver to promptly report inability to take the ride. Reassign when necessary, or notify the student if no driver is available. Administrators can view and manually reassign trips.

Define proximity, response timing, and safeguards against overlapping assignments during detailed design.

12. Admin dashboard

- Students: Review documents, approve or reject verification, request resubmission, and manage accounts and support cases.
- Drivers and vehicles: Manage profiles, vehicles and required documents; view duty status and active assignments.
- Rides and area: Monitor active and completed trips, reassign drivers, manage mapped boundaries and pickup points, and configure fares and trip limits.
- Wallets and support: Review relevant transactions, refunds, adjustments, disputes, incidents and support requests.
- Salary and attendance: Set salaries and pay periods, manage shifts, review attendance and work hours, and record or review salary payments. Drivers can see their own relevant records.

Agree how hours are calculated and corrected and how salaries are actually paid.

13. UI and UX direction

Give the student app and web app a consistent, youthful identity with bold typography, rounded cards, an energetic accent color and subtle animation. Keep booking, fares and safety clear.

The home screen can include a prominent “Where to?” field, saved destinations and campus pickup points, a service-area map, a wallet balance with Add money, and a visible Safety action. Support dark mode, readable contrast, accessible tap targets, screen readers and reduced motion.

The web app should support core student account management, booking, wallet information and trip history. Confirm first-release capabilities during planning.

14. Suggested first release

Prioritize student registration and verification; configurable service area and trip limit; fares and wallet payments; booking and automatic assignment; driver trip handling, PIN and live tracking; free cancellations and applicable refunds; administrative management of drivers, trips, salary and attendance; and essential safety and support.

15. Open decisions and pre-launch checks

- Approve the exact map boundary and verify road distances against the 18 km limit.
- Choose wallet payment methods and provider; decide unused-balance withdrawal and payment handling.
- Define shared-ride matching, base fares, capacity and cancellation effects.
- Confirm operating hours, initial driver and vehicle capacity, scheduling, attendance corrections and salary payment.
- Define verification review, document retention and access, support, emergency and partial-trip procedures.
- Choose the product name, branding and visual style.